

EE3621: Digital Signal Processing

Student Lecture Notes — Lecture 25

Design of IIR Filters — Bilinear Transformation & Matched Z-Transform

III B.Tech EEE — Core Exam Preparation Guide

1 Core Concept of the Bilinear Transformation (BLT)

1.1 Definition & Conformal Mapping

The **Bilinear Transformation** is an algebraic conformal mapping derived from the trapezoidal numerical integration rule. It converts an analog filter $H_a(s)$ directly into a digital filter $H(z)$ by algebraic substitution:

$$s = \frac{2}{T} \left(\frac{1 - z^{-1}}{1 + z^{-1}} \right) = \frac{2}{T} \left(\frac{z - 1}{z + 1} \right) \quad (1)$$

where T is the sampling interval (often chosen as $T = 1$ s or $T = 2$ s if unspecified).

1.2 Stability & The Elimination of Aliasing

- **Guaranteed Stability:** The entire open left-half of the s -plane ($\text{Re}(s) < 0$) maps strictly to the interior of the unit circle ($|z| < 1$). Stable analog filters **always produce stable digital filters**.
- **Zero Aliasing:** The entire continuous imaginary axis $s = j\Omega$ from $-\infty$ to $+\infty$ maps ****one-to-one**** onto the discrete unit circle $z = e^{j\omega}$ from $-\pi$ to $+\pi$.
- **Universal Applicability:** Because there is no aliasing, BLT can design ****Lowpass, Highpass, Bandpass, and Bandstop filters****!

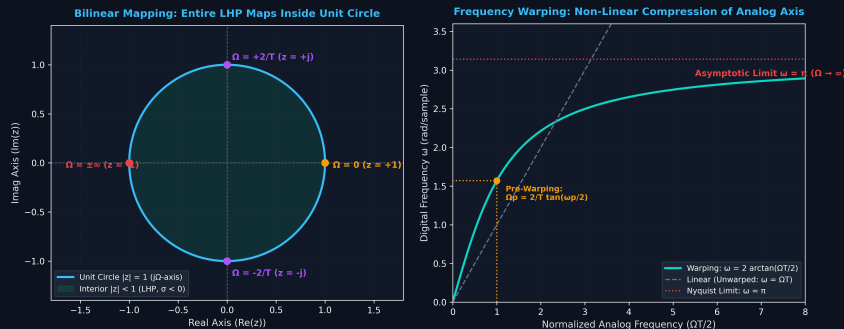


Figure 1: Left: One-to-one mapping from s -plane to z -plane. Right: Non-linear frequency warping curve showing compression into the digital interval $[0, \pi]$.

2 Frequency Warping & Mandatory Pre-Warping

2.1 The Frequency Warping Equation

Evaluating $s = \frac{2}{T} \left(\frac{1-e^{-j\omega}}{1+e^{-j\omega}} \right)$ yields the non-linear relationship between analog frequency Ω and digital frequency ω :

$$\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right) \iff \omega = 2 \arctan\left(\frac{\Omega T}{2}\right) \quad (2)$$

The infinite analog axis $[0, \infty)$ is non-linearly compressed into the finite digital interval $[0, \pi)$.

2.2 The Pre-Warping Rule

To ensure that the digital filter has its critical cutoff frequencies at the exact desired positions ω_p and ω_s , the designer **must pre-warp** the digital frequencies into analog prototype frequencies:

$$\Omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right), \quad \Omega_s = \frac{2}{T} \tan\left(\frac{\omega_s}{2}\right) \quad (3)$$

3 The Matched Z-Transform (MZT) Method

The **Matched Z-Transform** maps poles and zeros individually via direct exponential substitution:

$$(s - s_k) \longrightarrow (1 - e^{s_k T} z^{-1}) \quad (4)$$

- **Poles:** $s = p_k \implies z_k = e^{p_k T}$.
- **Zeros at Infinity:** For an all-pole analog filter with N poles and no finite zeros, add $(1+z^{-1})$ factors in the numerator to ensure transmission nulls at Nyquist ($\omega = \pi, z = -1$):

$$H(z) = K_d \frac{(1+z^{-1})^M}{\prod_{k=1}^N (1 - e^{p_k T} z^{-1})} \quad (5)$$

- **Gain Constant K_d :** Evaluated to match DC gain: $H(1) = H_a(0)$.

4 Master Comparison Table of IIR Design Methods

Table 1: Comparison of Impulse Invariance, Bilinear Transformation, and Matched Z-Transform

Feature	Impulse Invariance	Bilinear Transformation (BLT)	Matched Z-Transform (MZT)
Substitution Rule	Partial fractions: $\frac{A}{s-s_k} \rightarrow \frac{TA}{1-e^{s_k T} z^{-1}}$	$s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$ into $H_a(s)$	$(s-a) \rightarrow (1-e^{aT} z^{-1})$
Aliasing	**Severe aliasing**	**Zero aliasing**	Mild aliasing
Pre-Warping	Not applicable	**Mandatory** ($\Omega = \frac{2}{T} \tan(\omega/2)$)	Not applicable
Filter Types	LPF and BPF only (No HPF/BSF)	**Universal (LPF, HPF, BPF, BSF)**	LPF and BPF primarily
Stability	Strictly preserved	Strictly preserved	Strictly preserved

5 Standard Solved Exam Example

Problem: Design a 2nd-order digital Butterworth lowpass filter with a 3-dB cutoff frequency $\omega_c = 0.2\pi$ rad/sample using the Bilinear Transformation with $T = 1$ second.

Solution:

Step 1: Pre-Warp the Digital Cutoff Frequency

$$\Omega_c = \frac{2}{T} \tan\left(\frac{\omega_c}{2}\right) = 2 \tan\left(\frac{0.2\pi}{2}\right) = 2 \tan(0.1\pi) = 2 \tan(18^\circ) \quad (6)$$

Since $\tan(18^\circ) \approx 0.32492$:

$$\Omega_c = 2 \times 0.32492 = 0.6498 \text{ rad/s} \quad (7)$$

Step 2: Normalized 2nd-Order Butterworth Analog Prototype A 2nd-order Butterworth lowpass prototype has the transfer function:

$$H_a(s) = \frac{\Omega_c^2}{s^2 + \sqrt{2}\Omega_c s + \Omega_c^2} \quad (8)$$

Substituting $\Omega_c = 0.6498$:

$$\Omega_c^2 = (0.6498)^2 \approx 0.4223, \quad \sqrt{2}\Omega_c = \sqrt{2}(0.6498) \approx 0.9190 \quad (9)$$

$$H_a(s) = \frac{0.4223}{s^2 + 0.9190s + 0.4223} \quad (10)$$

Step 3: Apply the Bilinear Transformation ($T = 1$ s) Substitute $s = 2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$:

$$H(z) = \frac{0.4223}{\left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + 0.9190 \left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right] + 0.4223} \quad (11)$$

Multiply numerator and denominator by $(1+z^{-1})^2$:

$$\text{Numerator} = 0.4223(1 + 2z^{-1} + z^{-2}) \quad (12)$$

$$\begin{aligned} \text{Denominator} &= 4(1 - 2z^{-1} + z^{-2}) + 1.8380(1 - z^{-2}) + 0.4223(1 + 2z^{-1} + z^{-2}) \\ &= (4 + 1.8380 + 0.4223) + (-8 + 0.8446)z^{-1} + (4 - 1.8380 + 0.4223)z^{-2} \\ &= 6.2603 - 7.1554z^{-1} + 2.5843z^{-2} \end{aligned} \quad (13)$$

Dividing throughout by 6.2603:

$$\mathbf{H}(z) = \frac{0.0675(1 + 2z^{-1} + z^{-2})}{1 - 1.1430z^{-1} + 0.4128z^{-2}} \quad (14)$$

Check: At $z = 1$ (DC), $H(1) = \frac{0.0675(4)}{1-1.1430+0.4128} = \frac{0.2698}{0.2698} = 1.0$.

6 Essential Exam Checklist & Traps

- **Pre-Warping First:** Never use $\Omega_c = \omega_c/T$ in BLT. Always pre-warp: $\Omega_c = \frac{2}{T} \tan(\omega_c/2)$.
- **Cross-Term in Denominator:** $(1 - z^{-1})(1 + z^{-1}) = 1 - z^{-2}$. Do not introduce an erroneous z^{-1} cross-term!
- **Stability Check:** The constant term in the monic denominator ($a_2 = 0.4128$) represents $|z_1 z_2| = r^2$. Since $r = \sqrt{0.4128} \approx 0.64 < 1$, poles are stable.
- **Zero at Nyquist:** All lowpass filters designed by BLT possess transmission zeros at $z = -1$ ($\omega = \pi$).